

The Aprecis 100

The AI Canon

100 papers that built artificial intelligence, from the first artificial neuron (1943) to the reasoning models of today.

Reference draft, v0.1
Aprecis, 2026

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Era 1: Foundations

1943 to 1985

1. A Logical Calculus of Ideas Immanent in Nervous Activity

McCulloch and Pitts, 1943

First mathematical model of a neuron. Proved that networks of artificial neurons could compute logical functions. The seed of every neural network that followed.

2. Computing Machinery and Intelligence

Turing, 1950

Defined the question (can machines think?) and the Turing Test. Set the field's north star for 70 years.

3. The Perceptron: A Probabilistic Model for Information Storage and Organization

Rosenblatt, 1958

First learning machine. Showed that weights could be adjusted automatically from data. The press called it "the embryo of an electronic computer that thinks."

4. Some Studies in Machine Learning Using the Game of Checkers

Samuel, 1959

First program to learn from self-play. Coined the term "machine learning."

5. Perceptrons: An Introduction to Computational Geometry

Minsky and Papert, 1969

Proved perceptrons cannot learn XOR. Killed neural-network funding and triggered the first AI winter. Cultural villain, vindicated 30 years later.

6. Beyond Regression: New Tools for Prediction and Analysis in the Behavioral Sciences

Werbos, 1974 (PhD thesis)

Invented backpropagation. Sat ignored for a decade.

7. Neural Networks and Physical Systems with Emergent Collective Computational Abilities

Hopfield, 1982

Showed networks could store memories like a physicist's system. Reignited mainstream interest in neural networks.

8. Learning Representations by Back-Propagating Errors

Rumelhart, Hinton, Williams, 1986

Popularized backpropagation. The shot heard around connectionism.

9. Backpropagation Applied to Handwritten Zip Code Recognition

LeCun et al., 1989

First convolutional network. Postal automation became the first commercial neural-net deployment.

10. Finding Structure in Time

Elman, 1990

Simple recurrent network. Showed neural nets could handle sequences with memory.

11. Support-Vector Networks

Cortes and Vapnik, 1995

Support Vector Machines. Dominated machine learning for a decade and sidelined neural networks again.

12. Long Short-Term Memory

Hochreiter and Schmidhuber, 1997

Solved the vanishing gradient problem. Powered every recurrent net for 20 years until transformers arrived.

13. Gradient-Based Learning Applied to Document Recognition (LeNet-5)

LeCun, Bottou, Bengio, Haffner, 1998

Defined the modern CNN architecture: conv, pool, conv, pool, fully connected.

14. The Anatomy of a Large-Scale Hypertextual Web Search Engine

Brin and Page, 1998

PageRank. Showed machine learning at web scale. Built Google.

15. Random Forests

Breiman, 2001

Ensemble methods. Won every Kaggle competition for a decade.

16. Greedy Function Approximation: A Gradient Boosting Machine

Friedman, 2001

Gradient boosting. Foundation of XGBoost. Still the king of tabular data.

17. A Neural Probabilistic Language Model

Bengio et al., 2003

First neural language model. Direct ancestor of GPT.

18. Reducing the Dimensionality of Data with Neural Networks

Hinton and Salakhutdinov, 2006

Deep Belief Networks. First time "deep" learning actually worked. Started the deep-learning revival.

Era 3: Pre-Deep-Learning Groundwork

2001 to 2014

19. ImageNet: A Large-Scale Hierarchical Image Database

Deng, Dong, Socher, Li, Li, Fei-Fei, 2009

The dataset that enabled everything. No ImageNet, no AlexNet, no deep-learning boom. Datasets matter more than algorithms.

20. ImageNet Classification with Deep Convolutional Neural Networks (AlexNet)

Krizhevsky, Sutskever, Hinton, 2012

The big-bang moment. A 10-point error reduction on ImageNet shocked the field. GPUs plus data plus depth equals revolution.

21. Improving Neural Networks by Preventing Co-adaptation of Feature Detectors (Dropout)

Hinton et al., 2012

Regularization that finally let deep networks generalize instead of overfitting.

22. Efficient Estimation of Word Representations in Vector Space (Word2Vec)

Mikolov et al., 2013

Words as vectors. "King minus Man plus Woman equals Queen." Defined modern NLP.

23. Adam: A Method for Stochastic Optimization

Kingma and Ba, 2014

The optimizer everyone still uses. One of the most cited papers in modern machine learning.

24. Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift

Ioffe and Szegedy, 2015

Made very deep networks trainable. Standard ingredient in nearly every architecture after.

25. GloVe: Global Vectors for Word Representation

Pennington, Socher, Manning, 2014

Alternative to Word2Vec built from co-occurrence statistics. Widely used in production NLP.

26. Auto-Encoding Variational Bayes (VAE)

Kingma and Welling, 2013

Probabilistic generative models with latent variables. Foundation of all latent-space thinking.

Era 4: Deep Learning Explosion

2014 to 2017

27. Sequence to Sequence Learning with Neural Networks

Sutskever, Vinyals, Le, 2014

Encoder-decoder architecture. Defined modern translation, summarization, and dialogue systems.

- 28. Neural Machine Translation by Jointly Learning to Align and Translate**
Bahdanau, Cho, Bengio, 2014
First attention mechanism. The conceptual seed of transformers.
- 29. Generative Adversarial Networks**
Goodfellow et al., 2014
Two networks fighting. Launched the generative-modeling era. Yann LeCun called it "the coolest idea in machine learning in 20 years."
- 30. Going Deeper with Convolutions (GoogLeNet, Inception)**
Szegedy et al., 2014
Showed clever architecture beats raw depth. Inception modules became a standard pattern.
- 31. Very Deep Convolutional Networks for Large-Scale Image Recognition (VGG)**
Simonyan and Zisserman, 2014
Proved depth alone could help. Simple, hugely influential, easy to teach.
- 32. Deep Residual Learning for Image Recognition (ResNet)**
He, Zhang, Ren, Sun, 2015
Skip connections enabled 1000-plus-layer networks. The most cited paper of the 2010s.
- 33. Playing Atari with Deep Reinforcement Learning (DQN)**
Mnih et al., 2013
Deep RL beat humans at Atari games. Started DeepMind's rise to prominence.
- 34. Mastering the Game of Go with Deep Neural Networks and Tree Search (AlphaGo)**
Silver et al., 2016
Beat Lee Sedol 4 to 1. A cultural earthquake in Asia. Convinced governments that AI was real.
- 35. U-Net: Convolutional Networks for Biomedical Image Segmentation**
Ronneberger, Fischer, Brox, 2015
Standard for biomedical segmentation. Underpins the architecture of every modern diffusion model.
- 36. Fast R-CNN**
Girshick, 2015
Defined modern object detection.
- 37. You Only Look Once: Unified, Real-Time Object Detection (YOLO)**
Redmon, Divvala, Girshick, Farhadi, 2015
Real-time object detection. Spawned an industry of computer-vision startups.
- 38. A Neural Algorithm of Artistic Style**
Gatys, Ecker, Bethge, 2015
Neural style transfer. The first viral consumer AI moment.

39. FaceNet: A Unified Embedding for Face Recognition and Clustering

Schroff, Kalenichenko, Philbin, 2015

Face recognition at human level. Triplet loss became standard.

40. WaveNet: A Generative Model for Raw Audio

van den Oord et al., 2016

Raw audio generation. Powered Google Assistant voice.

41. Unsupervised Representation Learning with Deep Convolutional GANs (DCGAN)

Radford, Metz, Chintala, 2015

Made GANs actually trainable. Launched the generative-image wave.

42. Mask R-CNN

He, Gkioxari, Dollar, Girshick, 2017

Instance segmentation standard.

Era 5: Transformer Era

2017 to 2019

43. Attention Is All You Need

Vaswani et al., 2017

The transformer. Arguably the single most important paper of modern AI. Killed recurrent nets and birthed GPT, BERT, and everything since.

44. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Devlin, Chang, Lee, Toutanova, 2018

Bidirectional pretraining. Dominated NLP benchmarks overnight. Google Search runs on it.

45. Improving Language Understanding by Generative Pre-Training (GPT-1)

Radford, Narasimhan, Salimans, Sutskever, 2018

First GPT. Proved generative pretraining works for NLP.

46. Language Models are Unsupervised Multitask Learners (GPT-2)

Radford et al., 2019

1.5 billion parameters. OpenAI initially refused to release the model as "too dangerous." A marketing masterstroke.

47. Deep Contextualized Word Representations (ELMo)

Peters et al., 2018

Contextual embeddings. The bridge between Word2Vec and BERT.

48. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer (T5)

Raffel et al., 2019

Text-to-text framework. Unified all NLP tasks under one model.

49. RoBERTa: A Robustly Optimized BERT Pretraining Approach

Liu et al., 2019

Showed BERT was undertrained. Recipe matters as much as architecture.

50. Mastering the Game of Go Without Human Knowledge (AlphaGo Zero)

Silver et al., 2017

Mastered Go with zero human data. Pure self-play.

51. A General Reinforcement Learning Algorithm that Masters Chess, Shogi, and Go through Self-Play (AlphaZero)

Silver et al., 2017

Same algorithm beats chess, shogi, and Go. The generality moment.

52. Proximal Policy Optimization Algorithms (PPO)

Schulman et al., 2017

The reinforcement-learning workhorse. Powers RLHF.

53. The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks

Frankle and Carbin, 2018

Sparse subnetworks exist that match full-network performance. Reshaped pruning research.

54. A Style-Based Generator Architecture for Generative Adversarial Networks (StyleGAN)

Karras, Laine, Aila, 2019

Photorealistic faces. "This Person Does Not Exist" became the public's first deepfake panic.

Era 6: Scale Era

2020 to 2022

55. Language Models are Few-Shot Learners (GPT-3)

Brown et al., 2020

175 billion parameters. In-context learning emerged. The "scale is all you need" moment that birthed the modern LLM industry.

56. Scaling Laws for Neural Language Models

Kaplan et al., 2020

Predictable loss versus compute curves. Made scale a science instead of a guess.

57. Training Compute-Optimal Large Language Models (Chinchilla)

Hoffmann et al., 2022

GPT-3 was undertrained. Reshaped how every lab trains LLMs.

58. Denoising Diffusion Probabilistic Models (DDPM)

Ho, Jain, Abbeel, 2020

Made diffusion practical. The foundation of every modern image model.

- 59. Learning Transferable Visual Models From Natural Language Supervision (CLIP)**
Radford et al., 2021
Joint image-text embedding. Powers DALL-E, Stable Diffusion, and modern image search.
- 60. Zero-Shot Text-to-Image Generation (DALL-E)**
Ramesh et al., 2021
Text-to-image at scale. The public's first jaw-drop at generative AI.
- 61. High-Resolution Image Synthesis with Latent Diffusion Models (Stable Diffusion)**
Rombach et al., 2022
Open-source DALL-E. Democratized image generation overnight.
- 62. Evaluating Large Language Models Trained on Code (Codex)**
Chen et al., 2021
Powered GitHub Copilot. The first "AI does my job" moment for software engineers.
- 63. Highly Accurate Protein Structure Prediction with AlphaFold**
Jumper et al., 2021
Solved a 50-year biology problem. Nobel-prize-worthy. Legitimized AI for science.
- 64. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (ViT)**
Dosovitskiy et al., 2020
Transformer beats CNNs at vision. Unified architectures across modalities.
- 65. Masked Autoencoders Are Scalable Vision Learners (MAE)**
He et al., 2021
Self-supervised vision at scale. Vision's BERT moment.
- 66. Training Language Models to Follow Instructions with Human Feedback (InstructGPT)**
Ouyang et al., 2022
RLHF. Made GPT actually useful. The direct precursor of ChatGPT.
- 67. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models**
Wei et al., 2022
"Think step by step." Free reasoning. Started prompt engineering as a discipline.
- 68. PaLM: Scaling Language Modeling with Pathways**
Chowdhery et al., 2022
540 billion parameters. Google's scale flag-plant.
- 69. Flamingo: A Visual Language Model for Few-Shot Learning**
Alayrac et al., 2022
Few-shot multimodal. Set the template for GPT-4V and modern vision-language models.

70. Robust Speech Recognition via Large-Scale Weak Supervision (Whisper)

Radford et al., 2022

Open-source ASR at human level. Killed Otter, Rev, and a generation of transcription startups overnight.

71. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (RAG)

Lewis et al., 2020

Combine retrieval and LLM. Powers most enterprise AI today.

72. Constitutional AI: Harmlessness from AI Feedback

Bai et al., 2022

Anthropic. RLAIIF (AI feedback in place of human feedback). The technique behind Claude.

Era 7: Modern Frontier

2023 to 2026

73. ChatGPT Launch and System Card

OpenAI, 2022 to 2023

Not a research paper but a moment. 100 million users in 2 months. The fastest consumer-tech adoption in history.

74. GPT-4 Technical Report

OpenAI, 2023

Multimodal. No architecture details revealed. Marked the closed-frontier era.

75. LLaMA: Open and Efficient Foundation Language Models

Touvron et al., 2023

Open weights at frontier-adjacent quality. The Meta versus OpenAI war began here.

76. LLaMA 2: Open Foundation and Fine-Tuned Chat Models

Touvron et al., 2023

Commercially usable, RLHF in the open. Sparked the open-source LLM ecosystem.

77. Mistral 7B

Jiang et al., 2023

Small models punching far above their weight. Europe's first frontier-AI moment.

78. Mixtral of Experts (Mixtral 8x7B)

Jiang et al., 2024

Open Mixture-of-Experts. Sparse compute went mainstream.

79. Mamba: Linear-Time Sequence Modeling with Selective State Spaces

Gu and Dao, 2023

A real challenger to transformers. State-space models went mainstream.

80. Direct Preference Optimization (DPO)

Rafailov et al., 2023

RLHF without RL. Simpler, often better. Largely replaced PPO for alignment.

81. Toolformer: Language Models Can Teach Themselves to Use Tools

Schick et al., 2023

LLMs self-learn to use tools. Birthed the agents era.

82. ReAct: Synergizing Reasoning and Acting in Language Models

Yao et al., 2022

Think-then-act loop. The agent-loop template every framework now uses.

83. Tree of Thoughts: Deliberate Problem Solving with Large Language Models

Yao et al., 2023

Search over reasoning trees. Foundation for o1-style reasoning models.

84. Learning to Reason with LLMs (o1)

OpenAI, 2024

Test-time compute. Reasoning models became a distinct product category.

85. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek, 2025

Open reasoning model from China. RL-from-scratch worked. Wiped 600 billion dollars off Nvidia in one day.

86. Sora Technical Report

OpenAI, 2024

Video diffusion transformers. Hollywood-panic moment.

87. Video Generation Models as World Simulators

Brooks et al., 2024

Frames video models as physics simulators. Opened a new research direction.

88. A Generalist Agent (Gato)

Reed et al., 2022

One model, 600 tasks. The generalist-agent thesis stated plainly.

89. Voyager: An Open-Ended Embodied Agent with Large Language Models

Wang et al., 2023

LLM lives in Minecraft, learns skills. Embodied-agent proof-of-concept.

90. SWE-bench: Can Language Models Resolve Real-World GitHub Issues?

Jimenez et al., 2023

Benchmark for AI fixing real GitHub issues. Defined modern agent evaluation.

91. Scaling Monosemanticity / Towards Monosemanticity

Anthropic, 2024

Mechanistic interpretability at scale. We can finally see what features LLMs learn internally.

92. In-Context Learning and Induction Heads

Olsson et al., 2022

Discovered the circuit behind in-context learning. Interpretability landmark.

93. Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training

Anthropic, 2024

Models can hide misaligned behavior through training. Alignment alarm bell.

94. Weak-to-Strong Generalization

OpenAI, 2023

Can weak supervisors align stronger models? Foundation of the superalignment agenda.

95. Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context

Google DeepMind, 2024

1 million plus token context. The long-context era began.

96. The Claude 3 Model Family

Anthropic, 2024

Constitutional AI at frontier scale. Vibes-based benchmarks (taste) became a thing.

97. The Llama 3 Herd of Models

Meta, 2024

70B and 405B parameters. Open weights at GPT-4 class. Cemented the open-frontier movement.

98. Grok and the Real-Time Data Moat

xAI, 2024

Real-time training on X data. Articulated the new data-moat thesis.

99. AlphaProof and AlphaGeometry 2: Olympiad-Level Mathematical Reasoning

DeepMind, 2024

Olympic-level math via neuro-symbolic reasoning. AI does formal proofs.

100. The Current Frontier (rotating slot)

Updated quarterly

Reserved for the most recent landmark when a user reaches the top of the canon. Updates four times a year. Marketing line: the canon is alive. Subscribers always finish at the actual frontier.